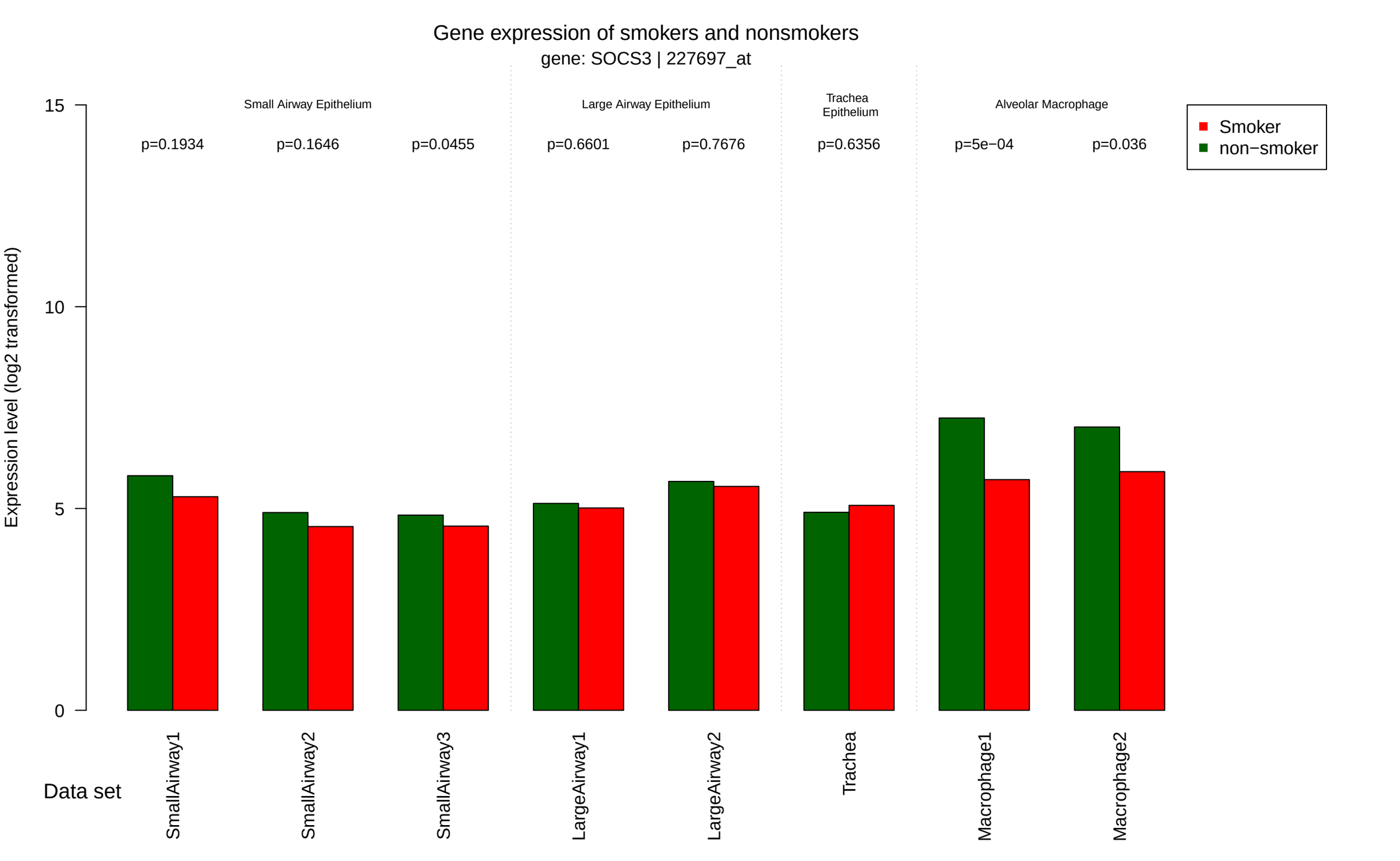
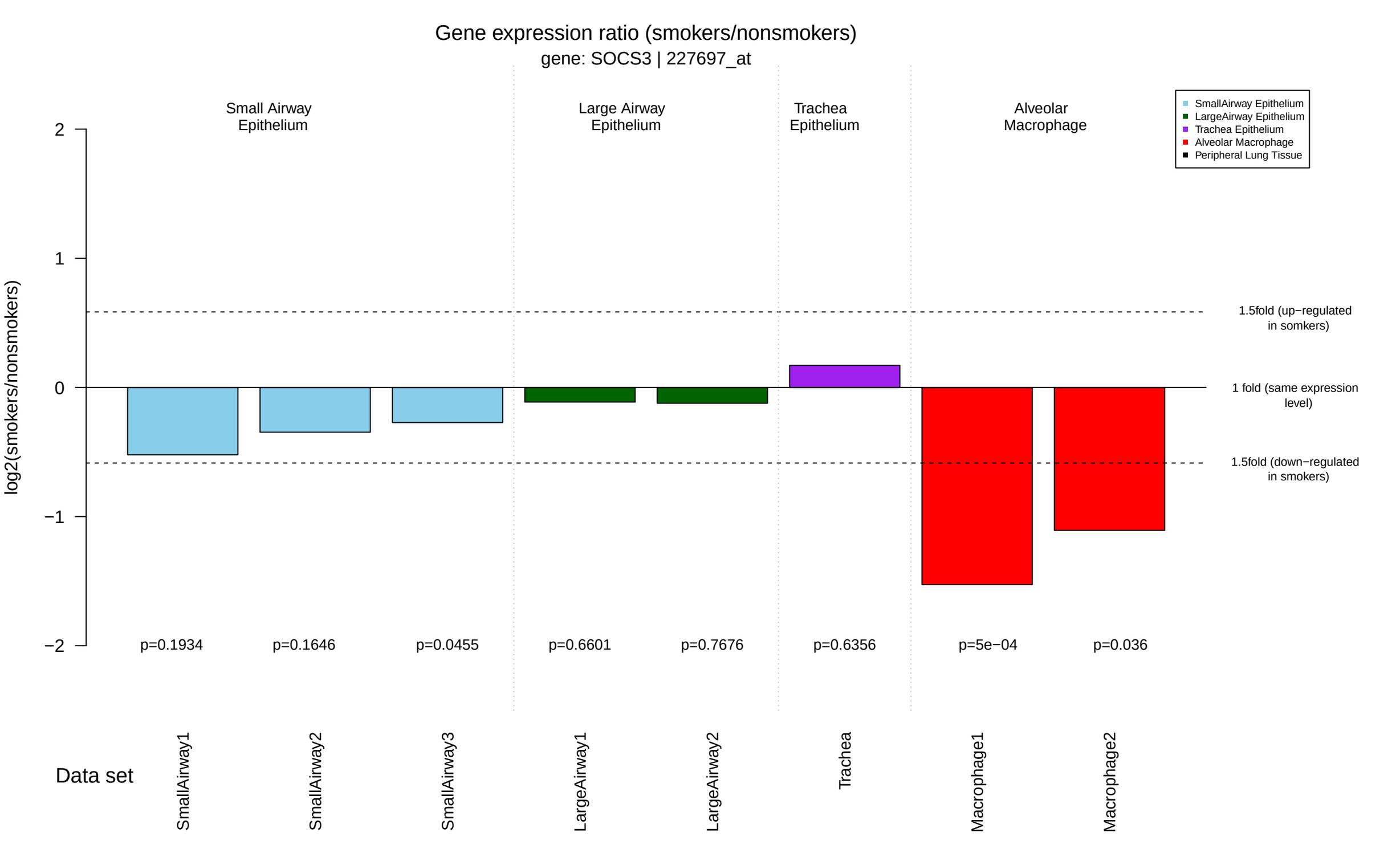
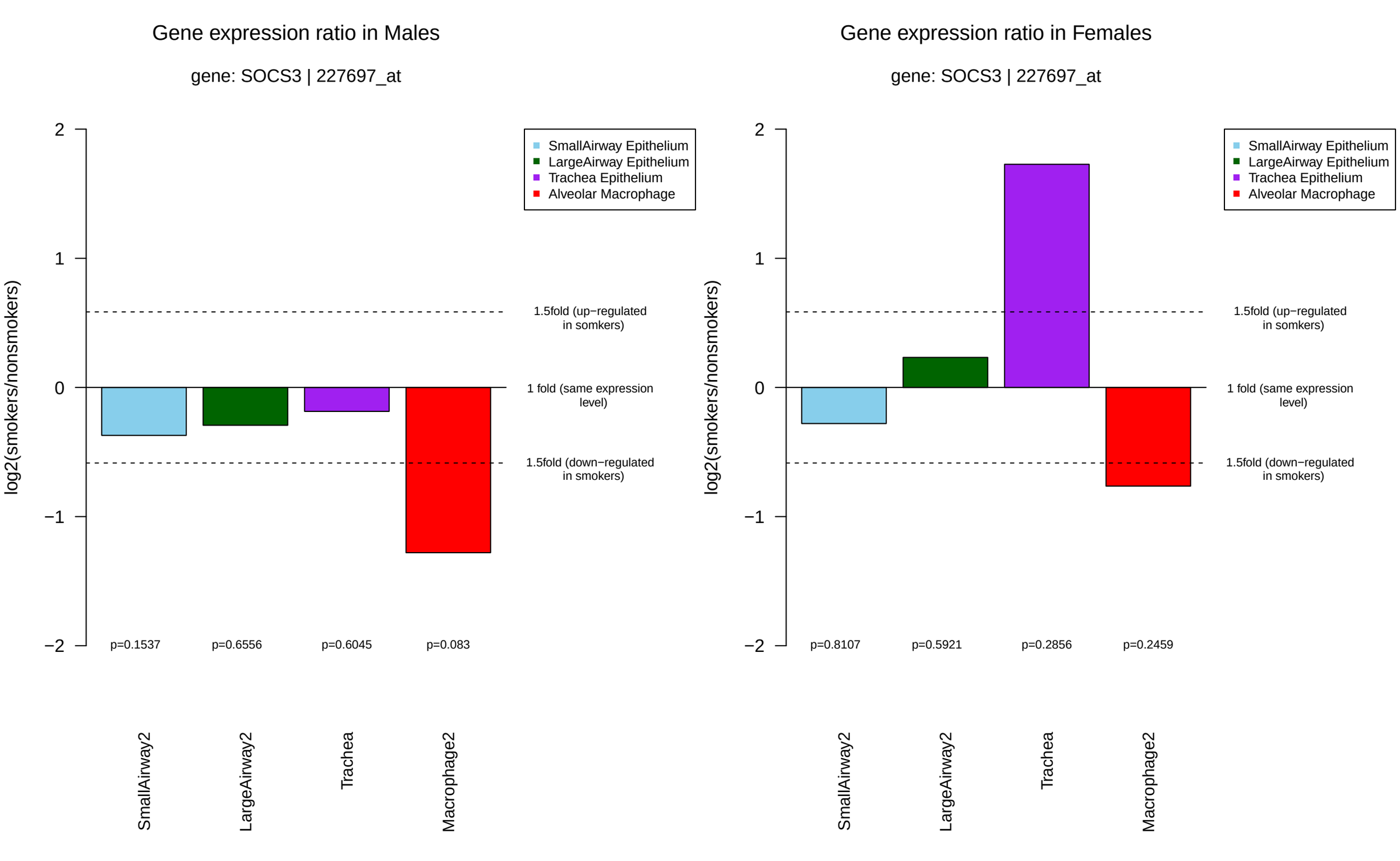
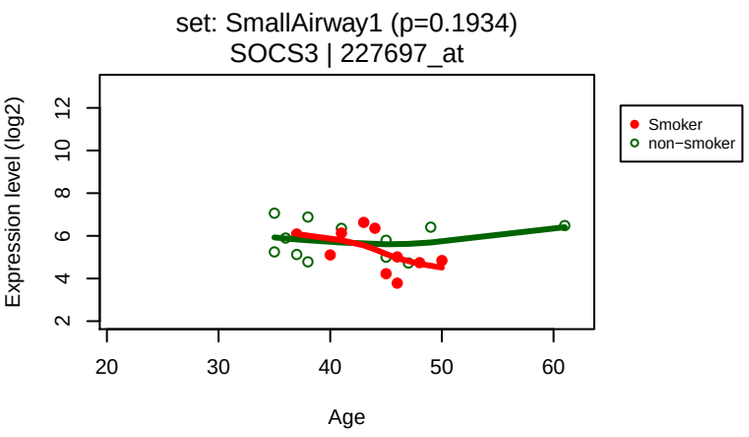
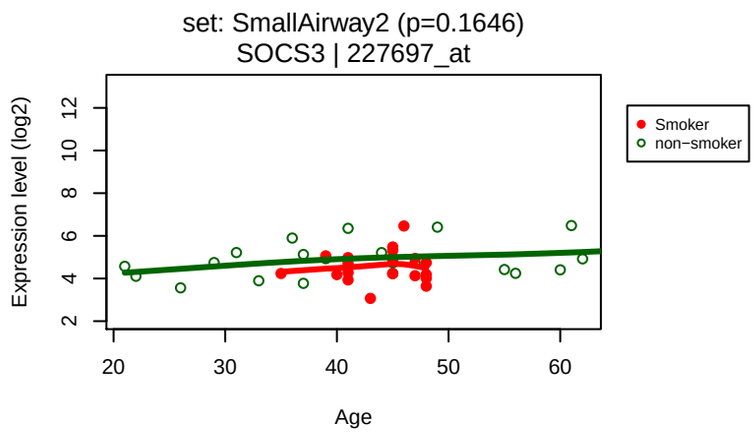
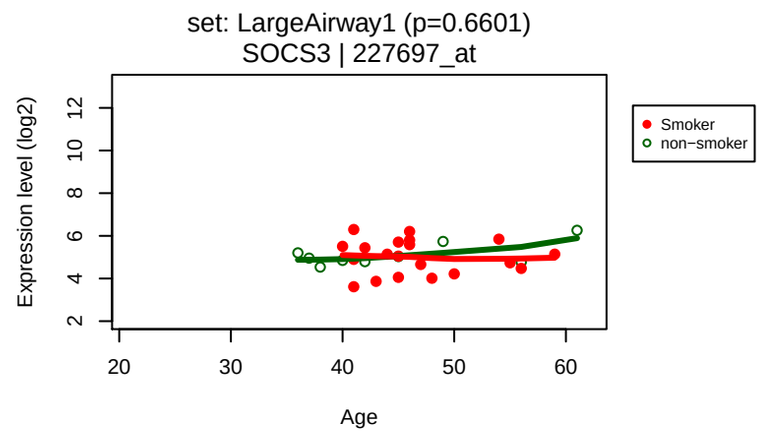

Gene expression of smokers and nonsmokers
gene: SOCS3 | 227697_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1934
p=0.1646
p=0.0455
p=0.6601
p=0.7676
p=0.6356
p=5e−04
p=0.036
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

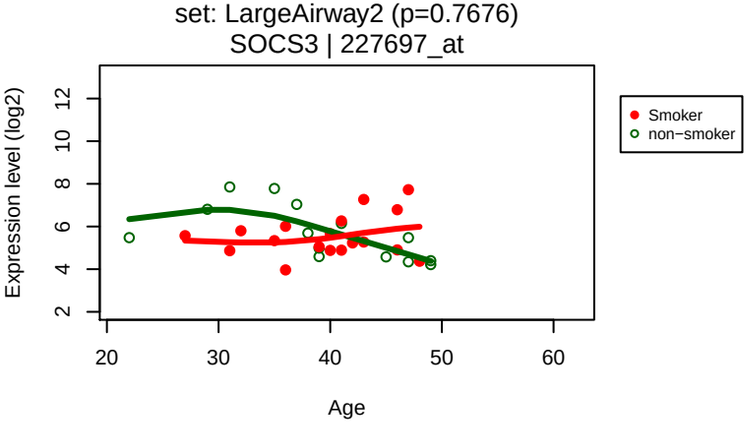

Gene expression ratio (smokers/nonsmokers)
gene: SOCS3 | 227697_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1934
p=0.1646
p=0.0455
p=0.6601
p=0.7676
p=0.6356
p=5e−04
p=0.036
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

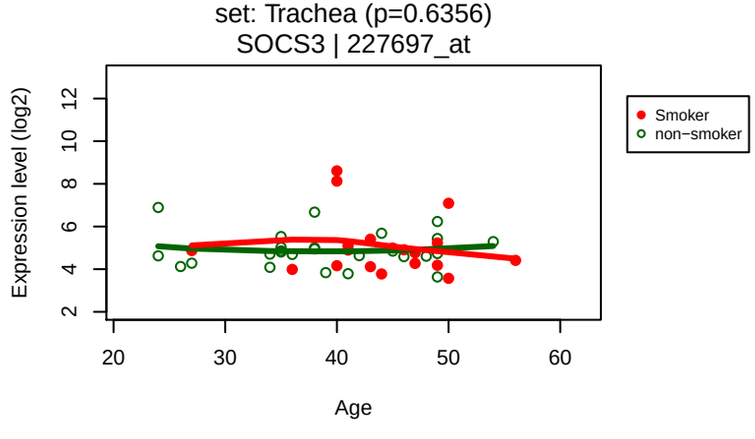

Gene expression ratio in Males
gene: SOCS3 | 227697_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1537
p=0.6556
p=0.6045
p=0.083
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SOCS3 | 227697_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8107
p=0.5921
p=0.2856
p=0.2459
SmallAirway2
LargeAirway2
Trachea
Macrophage2

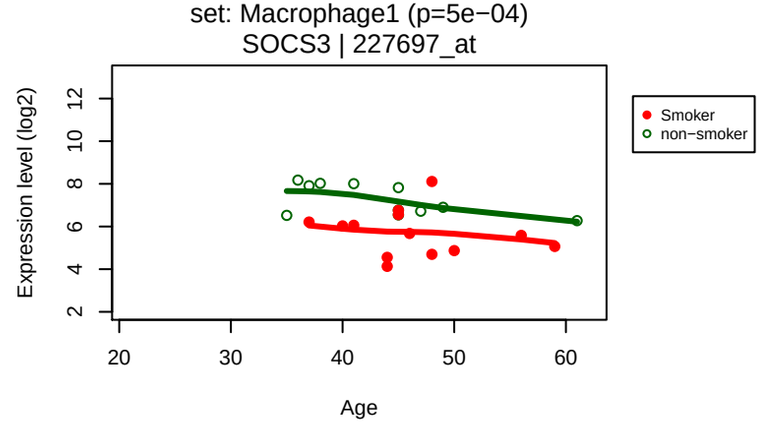

set: SmallAirway1 (p=0.1934)
SOCS3 | 227697_at
Expression level (log2)
Age
Smoker
non−smoker

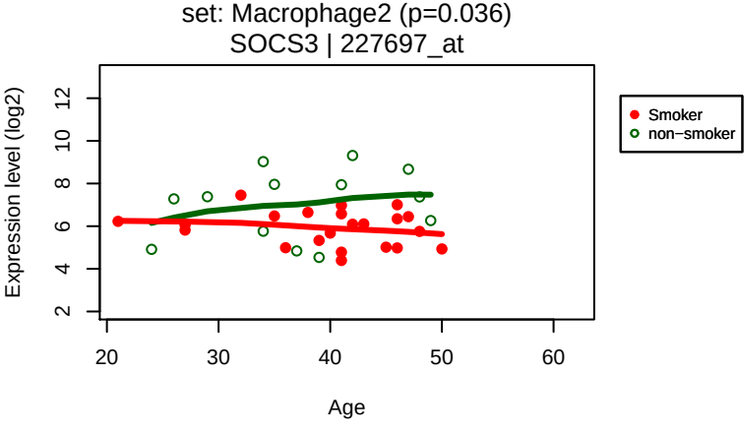

set: SmallAirway2 (p=0.1646)
SOCS3 | 227697_at
Expression level (log2)
Age
Smoker
non−smoker

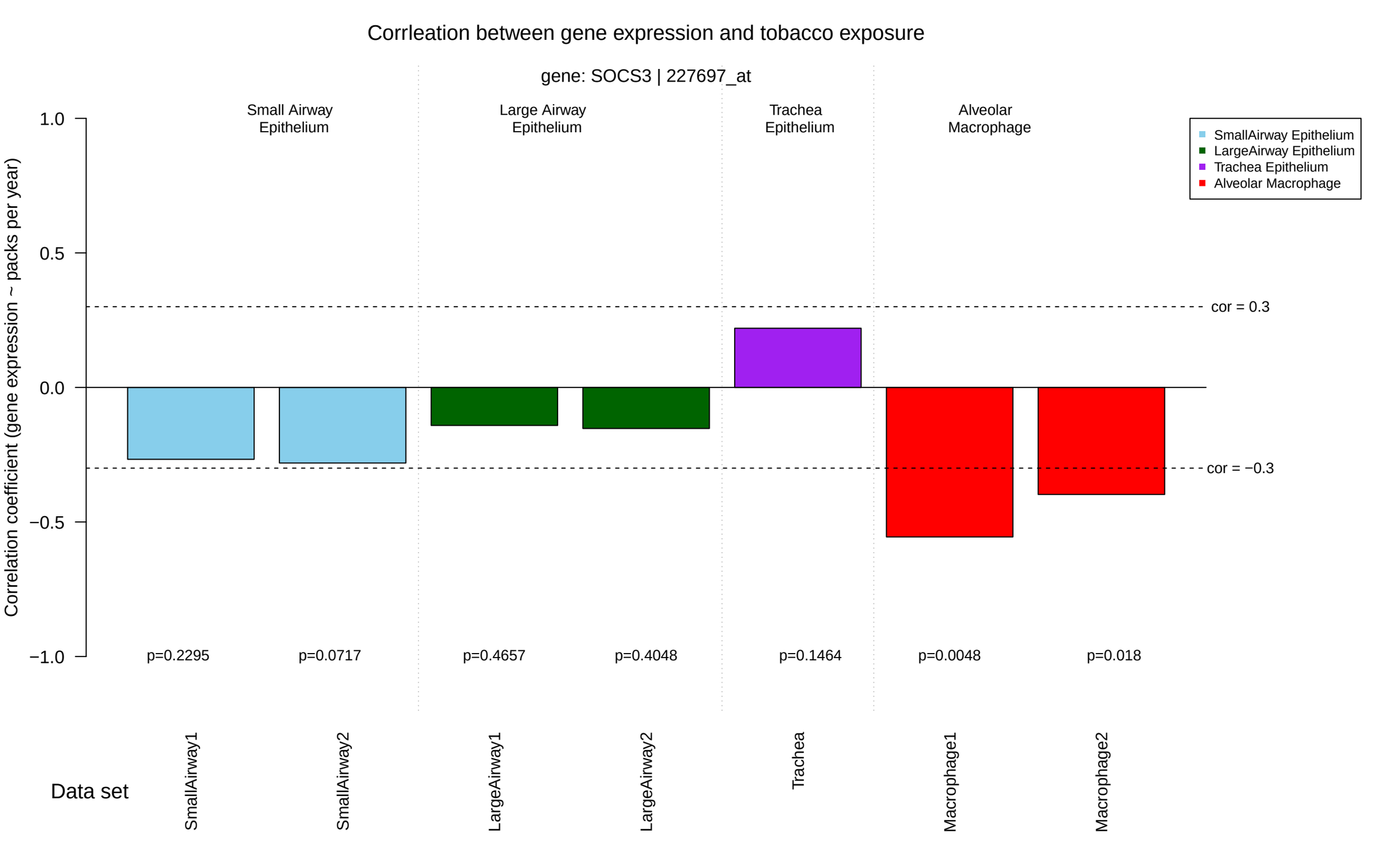

set: LargeAirway1 (p=0.6601)
SOCS3 | 227697_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7676)
SOCS3 | 227697_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6356)
SOCS3 | 227697_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=5e−04)
SOCS3 | 227697_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.036)
SOCS3 | 227697_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SOCS3 | 227697_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2295
p=0.0717
p=0.4657
p=0.4048
p=0.1464
p=0.0048
p=0.018
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2